

1. During which phase is the program loaded into memory?
 - a. Compile
 - b. Link
 - c. Load
 - d. Execute
2. In C++, what kind of files are produced by the compiler before the linking stage?
 - A) Executable files
 - B) Object files
 - C) Header files
 - D) Source files
3. When does a compiler or interpreter typically detect syntax errors in a program?
 - A) During execution
 - B) During debugging
 - C) During compilation
 - D) During planning
4. What is a key characteristic of compiled languages?
 - a. Code is executed line by line.
 - b. The entire source code is translated into machine code before execution.
 - c. Portability is a major advantage.
 - d. Debugging is typically done interactively.
5. Which programming language is often cited as an example of a compiled language?
 - a. Python
 - b. Java
 - c. C++
 - d. JavaScript
6. Which of the following accurately describes the purpose of a union in C++?
 - a. Groups related variables under a single name
 - b. Allows the creation of named integral constants
 - c. Shares the same memory location for all its members
 - d. Defines a collection of elements

7. Which of the following is a primary (fundamental) data type in C++?
- a. Array
 - b. Structure
 - c. Pointer
 - d. Long Double

8. What is the purpose of the following typedef declaration?

typedef unsigned long long int ULLI;

- a. It defines a new struct named ULLI.
 - b. It creates an alias for unsigned long long int called ULLI.
 - c. It declares a union named ULLI.
 - d. It specifies a set of named integer constants.
9. In which type does the **enumerators** are stored by the compiler?
- A. string
 - B. integer
 - C. float
 - D. none of the mentioned

10. What are the values of a, b, and c after this code executes?

```
int a = 5, b = 10, c;
```

```
c = a++ + --b
```

A) a=5, b=10, c=15

B) a=6, b=9, c=15

C) a=6, b=10, c=16

D) a=5, b=9, c=14

E) None of the mentioned

11. Which statement results from this code?

```
bool x = true, y = false;
```

```
bool z = x & y;
```

A) Error

B) z = true

C) z = false

D) None

12. In the switch statement, what is the purpose of the break keyword?
- a. To terminate the program.
 - b. To move to the next case.
 - c. To exit the loop.
 - d. To prevent fall-through to subsequent cases.
13. In a for loop, what happens if the update expression is omitted?
- a. It results in a compilation error.
 - b. The loop will not terminate.
 - c. The loop will not be executed
 - d. None
14. What happens if the condition of a do-while loop is always false?
- a. The loop executes once.
 - b. The loop does not execute.
 - c. The loop becomes an infinite loop.
 - d. The loop causes a compilation error
20. What are mandatory parts in the function declaration?
- a) return type, function name
 - b) return type, function name, parameters
 - c) parameters, function name
 - d) parameters, variables
21. How are many minimum numbers of functions need to be presented in c++?
- A. 0
 - B. 1
 - C. 2
 - D. 3
22. Function declaration is omitted when function is defined _____ main() function.
- A) Inside
 - B) Before
 - C) After
 - D) Anywhere

23. What is the purpose of the following code?

```
int main() {  
    int x;  
    cout << "Enter a number between 1 and 100: ";  
    while (true) {  
        cin >> x;  
        if (x >= 1 && x <= 100) {  
            break;  
        } else {  
            cout<< "Invalid input.."  
        }  
    }  
}
```

- a. It checks if the entered number is between 1 and 100 (inclusive) and continues execution.
- b. It continues to prompt the user until a valid number is entered.
- c. It prints an error message and exits if the entered number is not between 1 and 100.
- d. It ignores the user input and exits the program.

24. What is concept of function overloading in C++.

- A) Function overloading is a feature in C++ that allows defining multiple functions with the same name but different return types.
- B) Function overloading is a technique in C++ that allows defining multiple functions with the same name but different numbers or types of parameters.
- C) Function overloading is a practice in C++ that involves overwriting existing functions with new functionality.
- D) Function overloading is a way to create functions inside other functions to achieve better code organization.

25. What is the main purpose of a base case in a recursive function?

- a. To make the function call itself
- b. To provide an exit condition and stop the recursion
- c. To increase the efficiency of the recursive function
- d. To make the function tail-recursive

26. Which of the following is a potential issue with recursive solutions in terms of efficiency and memory usage?
- a. Recursive functions are always more memory-efficient.
 - b. Recursive functions consume additional memory for each function call, potentially leading to a stack overflow.
 - c. Recursive functions are faster than their iterative counterparts.
 - d. Recursive functions do not have any efficiency or memory-related concerns.
27. Where should default parameters appear in a function prototype?
- a) To the rightmost side of the parameter list
 - b) To the leftmost side of the parameter list
 - c) Anywhere inside the parameter list
 - d) Middle of the parameter list
28. By default, all the files are opened in _____ mode .
- a.** Binary
 - b.** Text
 - c.** Both
 - d. None
29. Which stream class is to only write on files?
- a) ofstream
 - b) ifstream
 - c) iostream
 - d) fstream
30. If we have object from ofstream class, then default mode of opening the file is ____ .
- a.** ios::in
 - b.** ios::out
 - c.** ios::in|ios::trunc
 - d.** ios::out|ios::trunc

31. Which of the following is not used to seek file pointer?

- a) ios::set
- b) ios::end
- c) ios::cur
- d) ios::beg

32. When using the seekp() function with ios::cur as the offset, what does it do?

- A. Sets the get pointer to the current position
- B. Moves the put pointer relative to the end of the file
- C. Moves the put pointer relative to the current position
- D. Sets the put pointer to the beginning of the file

33. What will be printed by this code segment?

```
int x = 5;
while (x <= 10) {
    if (x == 7)
        continue;
    cout << x << " ";
    x++;
}
```

- A) 5 6 7 8 9 10
- B) 5 6 8 9 10

- C) 5 6 8 9 10 11
- D) Infinite loop

E) none